

The background of the slide is a photograph of a wind farm at sunset or sunrise. The sky is a mix of deep blue and orange, with wispy clouds. Several wind turbines are visible in the distance, their silhouettes softened by the atmospheric haze. The overall mood is serene and modern, reflecting the renewable energy theme.

STRATEGIC DATA ANALYSIS

U.S. WIND FARM ACQUISITIONS

Exploratory Data Analysis Report

Introduction & Background

Wind energy is booming in the United States. As of 2021, there were over **70,000 wind turbines** spread across 46 states, generating over **140 GW** of power.

For a private equity firm looking to enter this space, simply buying any wind farm isn't enough. They need to understand the nuances of the market.

70,808

Total Turbines in Dataset

140+ GW

Installed U.S. Capacity

46

States with Turbines

27

Data Variables Analyzed



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Problem Statement & Project Goals

THE CHALLENGE

Our firm wants to acquire U.S. wind farms, but we need to know what drives **peak energy efficiency**.

We must understand what a historically successful wind farm looks like before investing.

THE GOAL

Identify the best **geographical regions**, **turbine specifications**, and **wind conditions** to guide our investments and maximize **ROI**.

Three Focus Areas



Geographic Focus

Find states with the most high-capacity turbines



Technology Track

Track how turbine specs have improved over the decades



Wind Resource Alignment

Combine turbine data with external wind speed and electricity cost data to find the best market opportunities

Data & Methodology

We analyzed 70,808 turbine records and merged with state-level wind speed and electricity pricing datasets.



U.S. Wind Turbine Database

70,808 turbine records from Lawrence Berkeley National Laboratory with 27 variables



Wind Speed Data

State-level average wind speeds across all 50 U.S. states



Electricity Pricing

State-level residential and commercial electricity rates



U.S. EIA-923 Power Plant Operations

Actual net generation (MWh) covering ~65% of the U.S. wind fleet.

ANALYSIS PIPELINE

01 Data Cleaning

Handle missing values, fix formatting errors, remove duplicates



02 Standardization

Map states to Census Regions, create decade buckets



03 External Merge

Merge wind speed and electricity pricing datasets



04 EDA & Insights

Exploratory analysis and strategic insights

The EDA Process

We analyzed 70,808 turbine records. This is the process we followed.

1

Auditing & Cleaning

Evaluated missing data (e.g., identifying that 91% of turbines have never been retrofitted) and fixed formatting errors.

2

Feature Engineering

Mapped states to U.S. Census Regions and grouped exact installation years into 10-year "Decade Buckets" for trend tracking.

3

Data Aggregation

Rolled up 70,000 individual turbine records into state-level summaries.

Capacity Factor Calculation:

$\text{Capacity Factor} = \text{Actual Energy Generated} / (\text{Nameplate Capacity} \times 8,760 \text{ hours})$

4

External Integration

Merged state-level wind speeds with our turbine data, categorizing continuous wind speeds into distinct "Wind Buckets" (Low, Moderate, High).

Agenda

Our roadmap for this strategic analysis

01

Geographic Concentration

Mapping turbine distribution across states and Census Regions

02

Technology Evolution

Tracking capacity, hub height, and rotor diameter improvements over time

03

Wind Resource Alignment

Cross-referencing wind speeds with turbine density and capacity

04

Recommendations

Actionable investment strategies for the acquisition committee

01

Geographic Concentration

Mapping turbine distribution across states and Census Regions

Geographic Dominance

KEY INSIGHTS

16,688

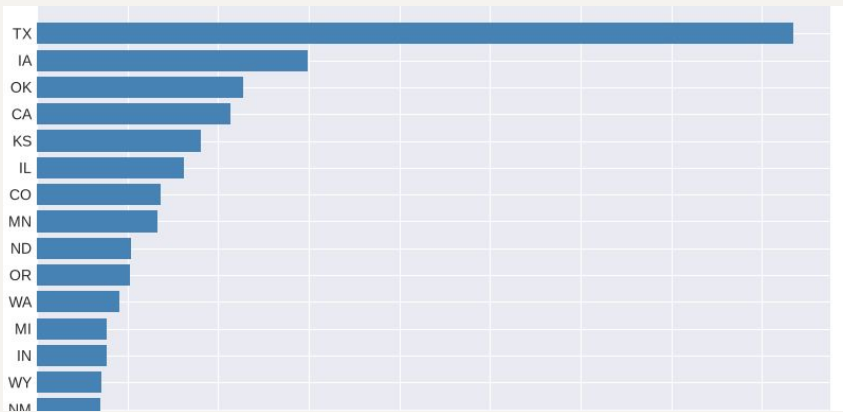
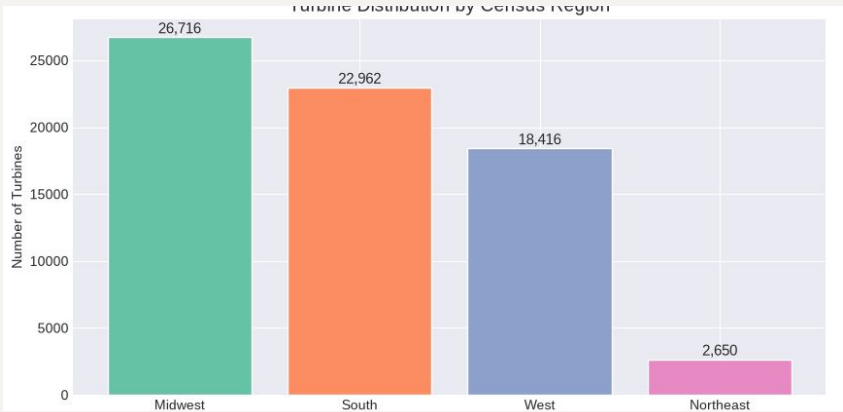
Texas turbines —almost the same as the entire Western U.S. combined

~26,700

Midwest turbines (38.5% of all U.S. installations)

55%

Top 5 states (TX, IA, OK, CA, KS) hold 55% of all U.S. turbines



02

Technology Evolution

Tracking capacity, hub height, and rotor diameter improvements over time

Technology Evolution

Average turbine capacity has grown **5x** since the 1990s

~500

kW avg capacity in 1990s



~3,000

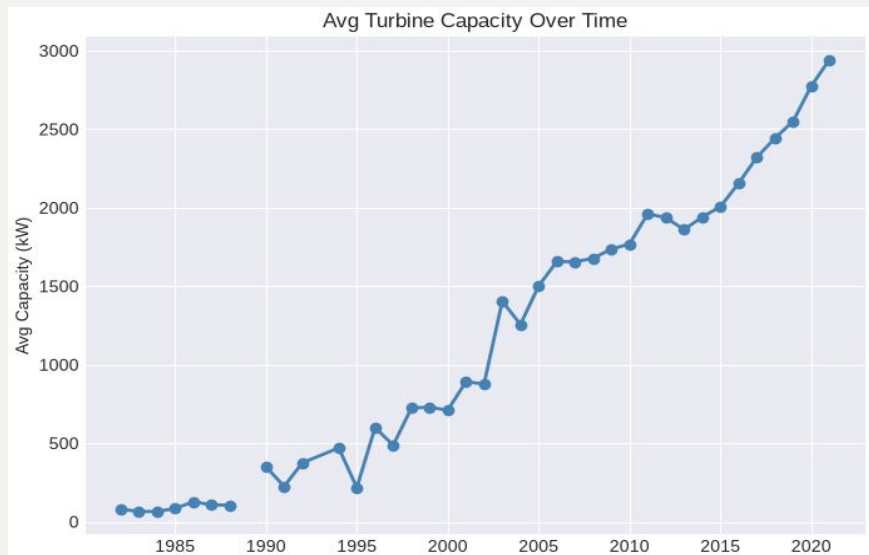
kW avg capacity by 2020

90m+

Modern hub height (vs. 50m in 1990s)

130m+

Modern rotor diameter (vs. 50m in 1990s)

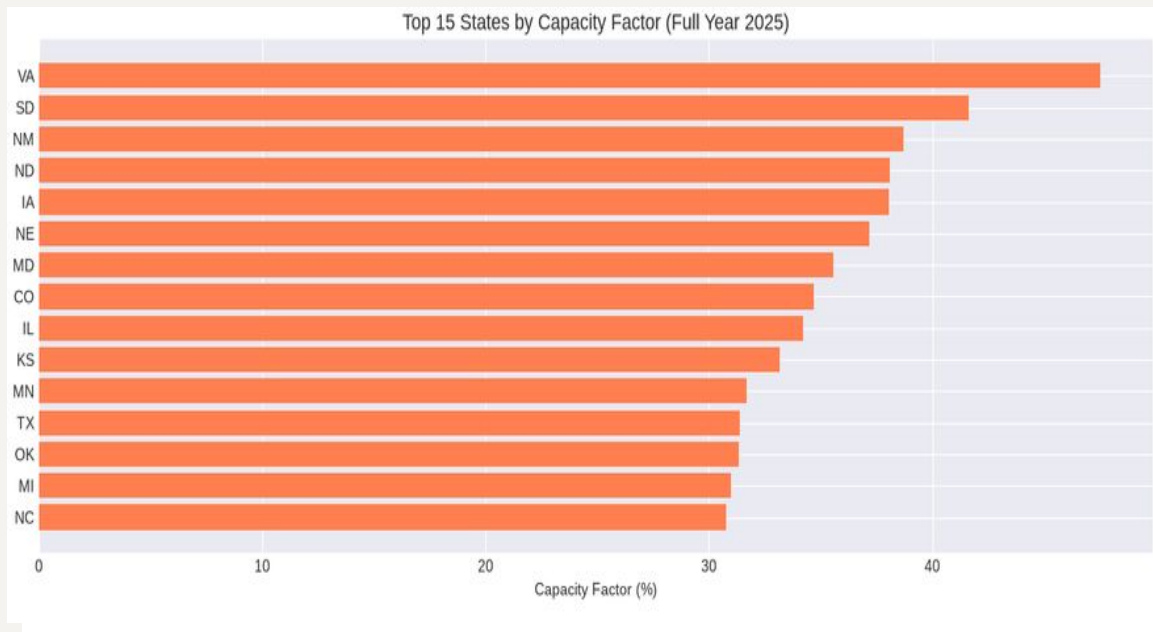


KEY TAKEAWAY

If we are buying older farms, we are buying **outdated tech.**

Newer is significantly better — post-2015 turbines average **>2,300 kW**.

The Reality of Capacity Factor



KEY INSIGHT

"High winds don't always equal highest efficiency."

Top States by Capacity Factor

VA — 47.5%

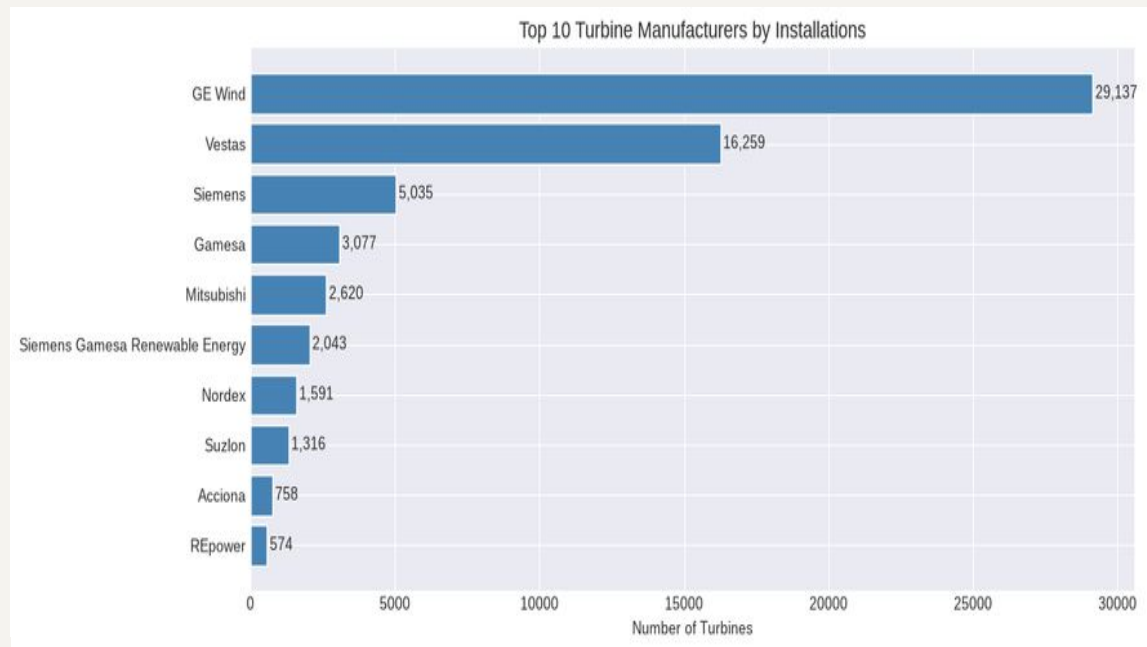
SD — 41.6%

NM — 38.7%

ND — 38.1%

IA — 38.1%

Manufacturer Dominance



KEY INSIGHT

"GE Wind supplies 41% of the U.S. market, followed by Vestas (23%)."

Market Share Breakdown

GE Wind — 41.1% (29,137 turbines)

Vestas — 23.0% (16,259 turbines)

Siemens — 7.1% (5,035 turbines)

Gamesa — 4.3% (3,077 turbines)

Siemens Gamesa — 2.9% (2,033 turbines)

Top 3 control ~71%

03

Wind Resource Alignment

Cross-referencing wind speeds with turbine density and capacity

The Wind "Sweet Spot"

Cross-referencing wind speeds with turbine installations

Wind Bucket	Turbine Count	Avg Capacity (kW)	Avg Hub Height (m)	Avg Rotor Diam (m)
Low (<14 mph)	4,290	1,361	63.1	66.8
Moderate (14–16)	20,188	2,037	81.3	97.9
Good (16–18)	13,143	2,053	83.1	98.2
High (18–20)	21,026	1,962	82.9	97.1
Very High (>20)	6,618	1,957	80.7	96.4

KEY INSIGHTS



Highest wind states (>20 mph: SD, MT, WY) are **underbuilt** — only 6,618 turbines



The **sweet spot** is "High" wind (18–20 mph) with 21,026 turbines



Low-wind states (<14 mph) show significantly lower capacity — **avoid** for acquisitions

Summary of Findings

01

Location is Highly Concentrated

Texas and the Midwest dominate the market with over **43%** of all U.S. turbines. Texas alone has **16,688** turbines.

02

Size Matters Enormously

Post-2015 turbines average **>2,300 kW** with 90m+ hub heights — exponentially more powerful than pre-2005 models.

03

Untapped Potential Exists

High-wind states (SD, MT, WY >20 mph) are **underbuilt**. The sweet spot is 18-20 mph range with optimal density.

5x

Capacity growth since 1990s

71%

Market share by top 3 manufacturers

18-20

mph optimal wind speed range

47.5%

Peak capacity factor (VA)

Actionable Recommendations

What the investment committee should do next



Target High Capacity Factor Regions (SD, NM)

Prioritize South Dakota and New Mexico where capacity factors exceed 38%, indicating highly efficient operations.



Focus on GE / Vestas Hardware

GE Wind and Vestas together control 64% of the market with proven track records in high-efficiency segments.



Target the Midwest

Prioritize acquisitions in Iowa, Illinois, and Minnesota where wind speeds and turbine density align optimally.



Focus on Modern Assets

Only acquire farms built post-2015 to ensure turbines producing upwards of 3 MW with 90m+ hub heights.

Data-Driven Investment Strategy: High CF Regions + Proven Hardware + Modern Assets = Maximized ROI

The background of the slide features a series of wind turbines in silhouette, receding into the distance. They are set against a sky with warm, orange and yellow hues from a low sun, with some wispy clouds. The overall tone is professional and clean.

THANK YOU

U.S. Wind Farm Acquisitions — Strategic Data Analysis